Operating Systems - Lab 8 (Mar 4)

Multithreading in Java – Using Thread class methods

1. Write a multithreaded program in Java that performs the following computation:

$$p = \sin(x) + \cos(y) + \tan(z)$$

This program should work as follows: On command line the user will enter values of x, y, z (in degrees). The program will then create three threads, one each for sin, cos and tan trigonometric function computation. When all the threads finish execution, the parent thread will perform the addition on the values generated by the child threads and output the final value.

2. The Fibonacci sequence is the series of numbers 0, 1, 1, 2, 3, 5, 8, ...
Formally, it can be expressed as:

$fib_0 = 0$
$fib_1 = 1$
$fib_n = fib_{n-1} + fib_{n-2}$

Write a multithreaded program in Java that generates the Fibonacci sequence. This program should work as follows: On the command line, the user will enter the number of Fibonacci numbers that the program is to generate. The program will then create a separate thread that will generate the Fibonacci numbers, placing the sequence in an array that can be shared by the threads. When the thread finishes execution, the parent thread will output the sequence generated by the child thread. Because the parent thread cannot begin outputting the Fibonacci sequence until the child thread finishes, the parent thread will have to wait for the child thread to finish.